INCH-POUND

MIL-DTL-32630/1
3 July 2019

DETAIL SPECIFICATION SHEET

CONDUCTOR, ELECTRICAL, STRANDED, UNINSULATED CARBON-BASED CONDUCTIVE FIBER, 0.014 NOMINAL DIAMETER, INCLUSIVE, -65 DEGREES CELSIUS to +150 DEGREES CELSIUS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32630.

REQUIREMENTS:

TABLE I. Conductor properties.

Part or Identifying Number (PIN)	Conductor type (see 1/)	Coating type (see 2/)	Tow (no. of filaments)	Filament diameter (in.)	Min. average break load (lb.) (see 3/)	Max. DC resistance (Ω/ft.)	Max. linear density (lbs./1,000 ft.)	Fiber CSA (in^2) (see 4/)
M32630/01-001	I	NC	178	0.00075	12	1.0	0.183	1.50E-04
M32630/01-002	I	SC	178	0.00076	12	1.0	0.185	1.54E-04

1/ Conductor type: Type I – Metal Coated Aramid Fiber (MCAF)
2/ Silver Coating (SC), Nickel Coating (NC),
3/ Minimum elongation of 1 percent
4/ Cross-section area (CSA) of metalized fiber

TABLE II. Size 26 conductor types.

Part or Identifying Number (PIN)	Conductor diameter (in.) (± .001)	Max. DC resistance (Ω/ft.)	Min. average break load (lbs.)	Min. fusing time (sec.) (at 3.25A)	Rated current (amperes)	Salt fog (hrs.)	Min. flex average life (cycles)
M32630/01-003	0.0180	0.67	15	90	1.25A	500	500
M32630/01-004	0.0180	0.67	15	90	1.25A	40	500

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The following requirements are specified for the table II size 26 conductors.

1. Conductor type: Type I, Metal Coated Aramid Fiber (MCAF)

2. Elongation:
   M32630/01-003 Minimum elongation of 1 percent.
   M32630/01-004 Minimum elongation of 2.5 percent.

3. Twists per inch: 2. When requested by the manufacturer, the qualifying activity can approve use of a different value.

4. Balance ply: Required.

5. Temperature coefficient of resistance shall be as specified in table III:

TABLE III. Temperature coefficient of resistance for size 26 conductor.

Part or Identifying Number (PIN)	Temperature coefficient of resistance ($\Delta\Omega/\Omega$@20°C)
M32630/01-003 Type I	0.003966
M32630/01-004 Type I	0.003966

NOTES:

1. Nominal metal percentage: 75 percent of the fiber weight

2. Typical weight savings compared to copper conductor: 75 percent

3. Intended use. This conductor is typically employed in EMI shield and multi-strand conductor applications.

4. The conductor is manufactured in bulk lengths (ten thousands to 100 thousands of feet per lot) and supplied on bobbins in lengths for further processing (hundreds to thousands of feet per bobbin) in braiding, weaving or multi-stranding operations.

5. Diameter (as-round): 0.012-0.016 inch when twisted to round under tension.
   Nominal flat width (as-braided): 0.025-0.035 inch.
   Nominal thickness (as-braided): 0.003-0.007 inch.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

(Project 6145-2018-051)

Review activities:
Army - AR
Navy - SH
Air Force - 11, 20

NOTE: The activities above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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